

Battery Charging



Special Tool(s)

	GR 1 190 V3.0 Intelligent Diagnostic Charger 162-00036 or equivalent
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Battery Charging Overview

NOTE: Batteries discharge while the vehicle is on the dealer lot or parked by the customer for an extended period of time due to normal parasitic key-off loads. Also, vehicles still in dealer inventory or in long-term storage may be driven short distances with heavy electrical loads. Over a period of time (30 days or more), this could result in vehicles having shallow or deeply discharged batteries.

The vehicle charging system is designed to supply the electrical power needed to maintain the battery near full charge during normal vehicle use. The charging system is not capable of bringing a deeply discharged battery back to near full charge in a short amount of time such as allowing the vehicle to idle for 15 minutes to "recharge the battery." Discharged batteries should be charged using an external charger.

NOTE: Battery chargers have improved greatly with the addition of the new generation pulse chargers. These chargers pulse current into the battery, breaking down the sulfation layer on the battery plates and generally reduce charging times to less than an hour.

NOTE: Cold batteries will not readily accept a charge. Therefore, batteries should be allowed to warm to approximately 5°C (41°F) before charging. This may require 4 to 8 hours at room temperature.

1. The following chart summarizes 2 recommended methods of charging.

Type of Battery Discharge	Pulse Charger	Standard Charger
Deeply discharged ^a	Follow directions supplied with the pulse charger	2 to 8 hours and may take up to an hour to accept initial charge
Shallow discharge ^b	45 minutes to an hour charge	2 hours below (40A) on manual setting or medium automatic setting

^aA deeply discharged is a battery that is drained over a prolonged period of time, such as an unsold vehicle or a vehicle in storage, to the point the battery is dead.

^bA shallow discharged battery is a battery that is drained by leaving an accessory on for several hours or a few days and has a very low charge.

NOTE: When charging the vehicle battery by connecting the charger to the negative battery terminal is necessary, such as when using a combination battery charger and battery tester/analyzer, like the GR 1 190 V3.0 Intelligent Diagnostic Charger, the Body Control Module (BCM) will not immediately update the battery state of charge. In this instance, after charging, you must CARRY OUT the Battery Monitoring System (BMS) Reset using the scan tool.

1. Connect the positive (red) charger clamp to the positive battery post.
2. Connect the negative (black) charger clamp to the negative battery post.
3. Charge the battery according to the battery charger owner's literature.
4. After charging is complete, carry out the Battery Monitoring System (BMS) Reset using the scan tool.

Charger Connected to Engine or Chassis Ground

1. Connect the positive (red) charger clamp to the positive battery post.
2. **NOTE:** The negative charger clamp must be connected to an unpainted chassis surface or a solid engine component such as a generator mount or engine lifting eye.

Connect the negative (black) charger clamp to an engine or chassis ground.

3. Charge the battery according to the battery charger owner's literature.